Untitled Project

1. Executive Summary

Total Floor Area	120.00 m2
Total Concrete Volume	72.34 m3
Total Steel Weight	4104.70 kg
Total Estimated Cost	INR 607,994.65

2. Design Basis

Design Codes & Standards:
- IS 456:2000 (Code of Practice for Plain and Reinforced Concrete)

Material Properties:
- Concrete: M25 (Density: 25 kN/m3, E = 5000sqrt(fck))
- Steel: Fe415 (Density: 78.5 kN/m3, E = 2x10^5 MPa)

Loading & Analysis:
- Design Load: User Input (Factored)
- Load Path: Tributary Area Method (Corner: 25%, Edge: 50%, Interior: 100% of bay)
- Beam Analysis: Simplified Simply Supported/Continuous assumptions.
(M = wL^2/8 or wL^2/10, Deflection checked against L/250)

Column Design:
- Short Column Assumption (Le/r < 12)
- Sizing Criteria: Axial Stress Check (Pu/Ag <= 0.4 fck)
Note: Steel capacity (0.67 fy Asc) is reserved as safety margin.

Foundation Design:
- Isolated Square Footings
- Area Required = Load * 1.1 / SBC (1.1 factor for self-weight)
- Depth Check: Punching Shear at d/2 distance (Limit: 0.25sqrt(fck))

Estimation Basis:
- Concrete Cost: INR 5000/m3
- Steel Cost: INR 60/kg
- Rebar Ratios: Columns 150kg/m3, Beams 120kg/m3, Footings 80kg/m3

3. Column Schedule (IS 456 Cl. 26.5.3)

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Status
C1_L0	L0	(0.00, 0.00)	208.7	300x450	PASS
C1_L1	L1	(0.00, 0.00)	71.1	300x450	PASS
C5_L0	L0	(0.00, 4.00)	107.2	300x450	PASS
C5_L1	L1	(0.00, 4.00)	53.6	300x450	PASS
C6_L0	L0	(0.00, 6.00)	136.9	300x450	PASS
C6_L1	L1	(0.00, 6.00)	57.1	300x450	PASS

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Status
C7_L0	L0	(0.00, 9.00)	107.2	300x450	PASS
C7_L1	L1	(0.00, 9.00)	53.6	300x450	PASS
C2_L0	L0	(0.00, 12.00)	193.5	300x450	PASS
C2_L1	L1	(0.00, 12.00)	63.5	300x450	PASS
C16_L0	L0	(3.75, 0.00)	137.6	300x450	PASS
C16_L1	L1	(3.75, 0.00)	68.8	300x450	PASS
C17_L0	L0	(3.75, 6.00)	168.1	300x450	PASS
C17_L1	L1	(3.75, 6.00)	84.0	300x450	PASS
C18_L0	L0	(3.75, 12.00)	107.2	300x450	PASS
C18_L1	L1	(3.75, 12.00)	53.6	300x450	PASS
C8_L0	L0	(7.50, 0.00)	117.3	300x450	PASS
C8_L1	L1	(7.50, 0.00)	58.7	300x450	PASS
C9_L0	L0	(7.50, 4.00)	213.1	300x450	PASS
C9_L1	L1	(7.50, 4.00)	95.2	300x450	PASS
C10_L0	L0	(7.50, 6.00)	187.7	300x450	PASS
C10_L1	L1	(7.50, 6.00)	82.5	300x450	PASS
C11_L0	L0	(7.50, 9.00)	213.1	300x450	PASS
C11_L1	L1	(7.50, 9.00)	95.2	300x450	PASS
C12_L0	L0	(7.50, 12.00)	91.9	300x450	PASS
C12_L1	L1	(7.50, 12.00)	46.0	300x450	PASS
C3_L0	L0	(10.00, 0.00)	188.4	300x450	PASS
C3_L1	L1	(10.00, 0.00)	60.9	300x450	PASS
C13_L0	L0	(10.00, 4.00)	91.7	300x450	PASS
C13_L1	L1	(10.00, 4.00)	42.1	300x450	PASS
C14_L0	L0	(10.00, 6.00)	66.5	300x450	PASS
C14_L1	L1	(10.00, 6.00)	33.3	300x450	PASS
C15_L0	L0	(10.00, 9.00)	91.7	300x450	PASS
C15_L1	L1	(10.00, 9.00)	42.1	300x450	PASS
C4_L0	L0	(10.00, 12.00)	178.2	300x450	PASS
C4_L1	L1	(10.00, 12.00)	55.9	300x450	PASS

4. Reinforcement Schedule (IS 456 Cl. 26.5)

Col ID	Main Bars	Links (Stirrups)
C1_L0	8-16#	8# @ 150 c/c
C1_L1	8-16#	8# @ 200 c/c
C5_L0	8-16#	8# @ 150 c/c

Col ID	Main Bars	Links (Stirrups)
C5_L1	8-16#	8# @ 200 c/c
C6_L0	8-16#	8# @ 150 c/c
C6_L1	8-16#	8# @ 200 c/c
C7_L0	8-16#	8# @ 150 c/c
C7_L1	8-16#	8# @ 200 c/c
C2_L0	8-16#	8# @ 150 c/c
C2_L1	8-16#	8# @ 200 c/c
C16_L0	8-16#	8# @ 150 c/c
C16_L1	8-16#	8# @ 200 c/c
C17_L0	8-16#	8# @ 150 c/c
C17_L1	8-16#	8# @ 200 c/c
C18_L0	8-16#	8# @ 150 c/c
C18_L1	8-16#	8# @ 200 c/c
C8_L0	8-16#	8# @ 150 c/c
C8_L1	8-16#	8# @ 200 c/c
C9_L0	8-16#	8# @ 150 c/c
C9_L1	8-16#	8# @ 200 c/c
C10_L0	8-16#	8# @ 150 c/c
C10_L1	8-16#	8# @ 200 c/c
C11_L0	8-16#	8# @ 150 c/c
C11_L1	8-16#	8# @ 200 c/c
C12_L0	8-16#	8# @ 150 c/c
C12_L1	8-16#	8# @ 200 c/c
C3_L0	8-16#	8# @ 150 c/c
C3_L1	8-16#	8# @ 200 c/c
C13_L0	8-16#	8# @ 150 c/c
C13_L1	8-16#	8# @ 200 c/c
C14_L0	8-16#	8# @ 150 c/c
C14_L1	8-16#	8# @ 200 c/c
C15_L0	8-16#	8# @ 150 c/c
C15_L1	8-16#	8# @ 200 c/c
C4_L0	8-16#	8# @ 150 c/c
C4_L1	8-16#	8# @ 200 c/c

5. Foundation Schedule (IS 456 Cl. 34)

Col ID	Load (kN)	Size (m)	Depth (mm)	Conc (m3)
C1_L0	208.7	1.50 x 1.50	300	0.67
C2_L0	193.5	1.50 x 1.50	300	0.67
C3_L0	188.4	1.50 x 1.50	300	0.67
C4_L0	178.2	1.50 x 1.50	300	0.67
C5_L0	107.2	1.50 x 1.50	300	0.67
C6_L0	136.9	1.50 x 1.50	300	0.67
C7_L0	107.2	1.50 x 1.50	300	0.67
C8_L0	117.3	1.50 x 1.50	300	0.67
C9_L0	213.1	1.50 x 1.50	300	0.67
C10_L0	187.7	1.50 x 1.50	300	0.67
C11_L0	213.1	1.50 x 1.50	300	0.67
C12_L0	91.9	1.50 x 1.50	300	0.67
C13_L0	91.7	1.50 x 1.50	300	0.67
C14_L0	66.5	1.50 x 1.50	300	0.67
C15_L0	91.7	1.50 x 1.50	300	0.67
C16_L0	137.6	1.50 x 1.50	300	0.67
C17_L0	168.1	1.50 x 1.50	300	0.67
C18_L0	107.2	1.50 x 1.50	300	0.67

6. Beam Schedule (IS 456 Cl. 26.5.1)

ID	Size (mm)	Top Steel	Bot Steel	Stirrups
AB_H_11	250x450	2-12#	5-20#	2L-8# @ 300 c/c
AB_H_12	250x450	2-12#	5-20#	2L-8# @ 300 c/c
AB_H_13	250x300	2-12#	3-12#	2L-8# @ 198 c/c
AB_H_14	250x350	2-12#	5-12#	2L-8# @ 236 c/c
AB_H_15	250x350	2-12#	5-16#	2L-8# @ 236 c/c
AB_H_16	250x300	2-12#	3-12#	2L-8# @ 198 c/c
AB_H_17	250x450	2-12#	5-20#	2L-8# @ 300 c/c
AB_H_18	250x450	2-12#	5-20#	2L-8# @ 300 c/c
AB_V_1	250x350	2-12#	4-16#	2L-8# @ 236 c/c
AB_V_10	250x300	2-12#	5-16#	2L-8# @ 198 c/c
AB_V_2	250x300	2-12#	4-12#	2L-8# @ 198 c/c
AB_V_3	250x300	2-12#	5-16#	2L-8# @ 198 c/c
AB_V_4	250x350	2-12#	5-12#	2L-8# @ 236 c/c
AB_V_5	250x300	2-12#	2-12#	2L-8# @ 198 c/c
AB_V_6	250x300	2-12#	6-12#	2L-8# @ 198 c/c

ID	Size (mm)	Top Steel	Bot Steel	Stirrups
AB_V_7	250x350	2-12#	5-12#	2L-8# @ 236 c/c
AB_V_8	250x350	2-12#	5-20#	2L-8# @ 236 c/c
AB_V_9	250x300	2-12#	5-12#	2L-8# @ 198 c/c

7. Slab Schedule

ID	Thk (mm)	Main Mesh	Dist Mesh
S1	175	10# @ 300 c/c	10# @ 300 c/c
S2	125	10# @ 300 c/c	10# @ 300 c/c
S3	150	10# @ 300 c/c	10# @ 300 c/c
S4	150	10# @ 300 c/c	10# @ 300 c/c
S5	175	10# @ 300 c/c	10# @ 300 c/c
S6	125	10# @ 300 c/c	10# @ 300 c/c
S7	150	10# @ 300 c/c	10# @ 300 c/c
S8	150	10# @ 300 c/c	10# @ 300 c/c
S9	130	10# @ 300 c/c	10# @ 300 c/c
S10	125	10# @ 300 c/c	10# @ 300 c/c
S11	130	10# @ 300 c/c	10# @ 300 c/c
S12	130	10# @ 300 c/c	10# @ 300 c/c

8. Masonry Wall Schedule

Wall ID	Start (x,y)	End (x,y)	Len (m)	Thk (mm)	Material	Load (kN/m)
W1	(0.0, 0.0)	(10.0, 0.0)	10.00	230	Brick	9.0
W2	(10.0, 0.0)	(10.0, 12.0)	12.00	230	Brick	9.0
W3	(10.0, 12.0)	(0.0, 12.0)	10.00	230	Brick	9.0
W4	(0.0, 12.0)	(0.0, 0.0)	12.00	230	Brick	9.0
W5	(7.5, 4.0)	(10.0, 4.0)	2.50	230	Brick	9.0
W6	(7.5, 4.0)	(7.5, 9.0)	5.00	230	Brick	9.0
W7	(7.5, 9.0)	(10.0, 9.0)	2.50	230	Brick	9.0
W8	(0.0, 6.0)	(7.5, 6.0)	7.50	230	Brick	9.0

9. Structural Layout Plan

Column Coordinates: C1_L0: (0.00, 0.00) C2_L0: (0.00, 12.00) C3_L0: (10.00, 0.00) C4_L0: (10.00, 12.00) C5_L0: (0.00, 4.00) C6_L0: (0.00, 6.00) C7_L0: (0.00, 9.00) C8_L0: (7.50, 0.00) C9_L0: (7.50, 4.00) C10_L0: (7.50, 6.00) C11_L0: (7.50, 9.00) C12_L0: (7.50, 12.00) C13_L0: (10.00, 4.00) C14_L0: (10.00, 6.00) C15_L0: (10.00, 9.00) C16_L0: (3.75, 0.00) C17_L0: (3.75, 6.00) C18_L0: (3.75, 12.00) C1_L1: (0.00, 0.00) C2_L1: (0.00, 12.00) C3_L1: (10.00, 0.00) C4_L1: (10.00, 12.00) C5_L1: (0.00, 4.00) C6_L1: (0.00, 6.00) C7_L1: (0.00, 9.00) C8_L1: (7.50, 0.00) C9_L1: (7.50, 4.00) C10_L1: (7.50, 6.00) C11_L1: (7.50, 9.00) C12_L1: (7.50, 12.00) C13_L1: (10.00, 4.00) C14_L1: (10.00, 6.00) C15_L1: (10.00, 9.00) C16_L1: (3.75, 0.00) C17_L1: (3.75, 6.00) C18_L1: (3.75, 12.00)

10. Integrated Steel Breakdown

Bar Diameter	Total Weight (kg)
8 mm	413.8
10 mm	2036.3
16 mm	1654.6
TOTAL	4104.7

10. Structural Audit Report

Member	Check	Design Val	Limit	Status
C1_L0	Column Axial Capacity	208.7	1781.2	PASS
C2_L0	Column Axial Capacity	193.5	1781.2	PASS
C3_L0	Column Axial Capacity	188.4	1781.2	PASS
C4_L0	Column Axial Capacity	178.2	1781.2	PASS
C5_L0	Column Axial Capacity	107.2	1781.2	PASS
C6_L0	Column Axial Capacity	136.9	1781.2	PASS
C7_L0	Column Axial Capacity	107.2	1781.2	PASS
C8_L0	Column Axial Capacity	117.3	1781.2	PASS
C9_L0	Column Axial Capacity	213.1	1781.2	PASS
C10_L0	Column Axial Capacity	187.7	1781.2	PASS
C11_L0	Column Axial Capacity	213.1	1781.2	PASS
C12_L0	Column Axial Capacity	91.9	1781.2	PASS
C13_L0	Column Axial Capacity	91.7	1781.2	PASS
C14_L0	Column Axial Capacity	66.5	1781.2	PASS
C15_L0	Column Axial Capacity	91.7	1781.2	PASS
C16_L0	Column Axial Capacity	137.6	1781.2	PASS
C17_L0	Column Axial Capacity	168.1	1781.2	PASS
C18_L0	Column Axial Capacity	107.2	1781.2	PASS
C1_L1	Column Axial Capacity	71.1	1781.2	PASS
C2_L1	Column Axial Capacity	63.5	1781.2	PASS
C3_L1	Column Axial Capacity	60.9	1781.2	PASS
C4_L1	Column Axial Capacity	55.9	1781.2	PASS
C5_L1	Column Axial Capacity	53.6	1781.2	PASS
C6_L1	Column Axial Capacity	57.1	1781.2	PASS
C7_L1	Column Axial Capacity	53.6	1781.2	PASS
C8_L1	Column Axial Capacity	58.7	1781.2	PASS
C9_L1	Column Axial Capacity	95.2	1781.2	PASS

Member	Check	Design Val	Limit	Status
C10_L1	Column Axial Capacity	82.5	1781.2	PASS
C11_L1	Column Axial Capacity	95.2	1781.2	PASS
C12_L1	Column Axial Capacity	46.0	1781.2	PASS
C13_L1	Column Axial Capacity	42.1	1781.2	PASS
C14_L1	Column Axial Capacity	33.3	1781.2	PASS
C15_L1	Column Axial Capacity	42.1	1781.2	PASS
C16_L1	Column Axial Capacity	68.8	1781.2	PASS
C17_L1	Column Axial Capacity	84.0	1781.2	PASS
C18_L1	Column Axial Capacity	53.6	1781.2	PASS
F-C1_L0	Bearing Pressure	68.0	200.0	PASS
F-C1_L0	Punching Shear	0.4	1.2	PASS
F-C2_L0	Bearing Pressure	63.1	200.0	PASS
F-C2_L0	Punching Shear	0.4	1.2	PASS
F-C3_L0	Bearing Pressure	61.4	200.0	PASS
F-C3_L0	Punching Shear	0.3	1.2	PASS
F-C4_L0	Bearing Pressure	58.1	200.0	PASS
F-C4_L0	Punching Shear	0.3	1.2	PASS
F-C5_L0	Bearing Pressure	34.9	200.0	PASS
F-C5_L0	Punching Shear	0.2	1.2	PASS
F-C6_L0	Bearing Pressure	44.6	200.0	PASS
F-C6_L0	Punching Shear	0.3	1.2	PASS
F-C7_L0	Bearing Pressure	34.9	200.0	PASS
F-C7_L0	Punching Shear	0.2	1.2	PASS
F-C8_L0	Bearing Pressure	38.2	200.0	PASS
F-C8_L0	Punching Shear	0.2	1.2	PASS
F-C9_L0	Bearing Pressure	69.5	200.0	PASS
F-C9_L0	Punching Shear	0.4	1.2	PASS
F-C10_L0	Bearing Pressure	61.2	200.0	PASS
F-C10_L0	Punching Shear	0.3	1.2	PASS
F-C11_L0	Bearing Pressure	69.5	200.0	PASS
F-C11_L0	Punching Shear	0.4	1.2	PASS
F-C12_L0	Bearing Pressure	30.0	200.0	PASS
F-C12_L0	Punching Shear	0.2	1.2	PASS
F-C13_L0	Bearing Pressure	29.9	200.0	PASS
F-C13_L0	Punching Shear	0.2	1.2	PASS
F-C14_L0	Bearing Pressure	21.7	200.0	PASS
F-C14_L0	Punching Shear	0.1	1.2	PASS

Member	Check	Design Val	Limit	Status
F-C15_L0	Bearing Pressure	29.9	200.0	PASS
F-C15_L0	Punching Shear	0.2	1.2	PASS
F-C16_L0	Bearing Pressure	44.9	200.0	PASS
F-C16_L0	Punching Shear	0.3	1.2	PASS
F-C17_L0	Bearing Pressure	54.8	200.0	PASS
F-C17_L0	Punching Shear	0.3	1.2	PASS
F-C18_L0	Bearing Pressure	34.9	200.0	PASS
F-C18_L0	Punching Shear	0.2	1.2	PASS
AB_V_1_L0	Beam Deflection (Serviceability)	6.6	16.0	PASS
AB_V_2_L0	Beam Deflection (Serviceability)	0.6	8.0	PASS
AB_V_3_L0	Beam Deflection (Serviceability)	0.6	8.0	PASS
AB_V_4_L0	Beam Deflection (Serviceability)	6.6	16.0	PASS
AB_V_5_L0	Beam Deflection (Serviceability)	0.6	8.0	PASS
AB_V_6_L0	Beam Deflection (Serviceability)	3.3	12.0	PASS
AB_V_7_L0	Beam Deflection (Serviceability)	6.6	16.0	PASS
AB_V_8_L0	Beam Deflection (Serviceability)	6.6	16.0	PASS
AB_V_9_L0	Beam Deflection (Serviceability)	0.0	4.0	PASS
AB_V_10_L0	Beam Deflection (Serviceability)	3.3	12.0	PASS
AB_H_11_L0	Beam Deflection (Serviceability)	7.8	20.0	PASS
AB_H_12_L0	Beam Deflection (Serviceability)	7.8	20.0	PASS
AB_H_13_L0	Beam Deflection (Serviceability)	1.6	10.0	PASS
AB_H_14_L0	Beam Deflection (Serviceability)	5.1	15.0	PASS
AB_H_15_L0	Beam Deflection (Serviceability)	5.1	15.0	PASS
AB_H_16_L0	Beam Deflection (Serviceability)	1.6	10.0	PASS
AB_H_17_L0	Beam Deflection (Serviceability)	7.8	20.0	PASS
AB_H_18_L0	Beam Deflection (Serviceability)	7.8	20.0	PASS
AB_V_1_L1	Beam Deflection (Serviceability)	6.6	16.0	PASS
AB_V_2_L1	Beam Deflection (Serviceability)	0.6	8.0	PASS
AB_V_3_L1	Beam Deflection (Serviceability)	0.6	8.0	PASS
AB_V_4_L1	Beam Deflection (Serviceability)	6.6	16.0	PASS
AB_V_5_L1	Beam Deflection (Serviceability)	0.6	8.0	PASS
AB_V_6_L1	Beam Deflection (Serviceability)	3.3	12.0	PASS
AB_V_7_L1	Beam Deflection (Serviceability)	6.6	16.0	PASS
AB_V_8_L1	Beam Deflection (Serviceability)	6.6	16.0	PASS
AB_V_9_L1	Beam Deflection (Serviceability)	0.0	4.0	PASS
AB_V_10_L1	Beam Deflection (Serviceability)	3.3	12.0	PASS
AB_H_11_L1	Beam Deflection (Serviceability)	7.8	20.0	PASS

Member	Check	Design Val	Limit	Status
AB_H_12_B1	Beam Deflection (Serviceability)	7.8	20.0	PASS
AB_H_13_B1	Beam Deflection (Serviceability)	1.6	10.0	PASS
AB_H_14_B1	Beam Deflection (Serviceability)	5.1	15.0	PASS
AB_H_15_B1	Beam Deflection (Serviceability)	5.1	15.0	PASS
AB_H_16_B1	Beam Deflection (Serviceability)	1.6	10.0	PASS
AB_H_17_B1	Beam Deflection (Serviceability)	7.8	20.0	PASS
AB_H_18_B1	Beam Deflection (Serviceability)	7.8	20.0	PASS

11. Sustainability Audit

Design Strategy: Standard (OPC)

Material	Quantity	Emission Factor	Total CO2e (kg)
Concrete	72.34 m3	300 kg/m3	21702.7
Steel	4105 kg	1.85 kg/kg	7593.7
TOTAL	-	-	29296.4

Wait! You can reduce your Carbon Footprint.
Refining the design to use **Fly Ash based Concrete (PPC)** instead of OPC can reduce concrete emissions by approximately 15-20%.
Switch "Use Green Concrete" ON in the dashboard to see savings. (Note: PPC is slower setting and has lower early strength, use only if confirmed by RMC supplier).

12. Serviceability Checks (Deflection)

Beam ID	Description	Actual (mm)	Limit (mm)	Status
AB_V_1_L0	Span/250 Check	6.624	16.00	PASS
AB_V_2_L0	Span/250 Check	0.648	8.00	PASS
AB_V_3_L0	Span/250 Check	0.648	8.00	PASS
AB_V_4_L0	Span/250 Check	6.624	16.00	PASS
AB_V_5_L0	Span/250 Check	0.648	8.00	PASS
AB_V_6_L0	Span/250 Check	3.281	12.00	PASS
AB_V_7_L0	Span/250 Check	6.624	16.00	PASS
AB_V_8_L0	Span/250 Check	6.624	16.00	PASS
AB_V_9_L0	Span/250 Check	0.041	4.00	PASS
AB_V_10_L0	Span/250 Check	3.281	12.00	PASS
AB_H_11_L0	Span/250 Check	7.823	20.00	PASS
AB_H_12_L0	Span/250 Check	7.823	20.00	PASS
AB_H_13_L0	Span/250 Check	1.582	10.00	PASS
AB_H_14_L0	Span/250 Check	5.117	15.00	PASS
AB_H_15_L0	Span/250 Check	5.117	15.00	PASS
AB_H_16_L0	Span/250 Check	1.582	10.00	PASS
AB_H_17_L0	Span/250 Check	7.823	20.00	PASS
AB_H_18_L0	Span/250 Check	7.823	20.00	PASS
AB_V_1_L1	Span/250 Check	6.624	16.00	PASS
AB_V_2_L1	Span/250 Check	0.648	8.00	PASS
AB_V_3_L1	Span/250 Check	0.648	8.00	PASS
AB_V_4_L1	Span/250 Check	6.624	16.00	PASS
AB_V_5_L1	Span/250 Check	0.648	8.00	PASS
AB_V_6_L1	Span/250 Check	3.281	12.00	PASS
AB_V_7_L1	Span/250 Check	6.624	16.00	PASS
AB_V_8_L1	Span/250 Check	6.624	16.00	PASS
AB_V_9_L1	Span/250 Check	0.041	4.00	PASS
AB_V_10_L1	Span/250 Check	3.281	12.00	PASS
AB_H_11_L1	Span/250 Check	7.823	20.00	PASS
AB_H_12_L1	Span/250 Check	7.823	20.00	PASS
...	...	...	...	...

13. Staircase Design Schedule (Dog-Legged)

No staircase selected for design.

Seismic Analysis (IS 1893:2016)

Parameter	Value
Seismic Zone	III

Zone Factor (Z)	0.16
Design Acceleration (Ah)	0.0400
Fundamental Period (Ta)	0.390 s
Base Shear (Vb)	74.7 kN
Strong Column–Weak Beam	REQUIRED
SCWB Status	PASS

Wind Analysis (IS 875 Part 3)

Parameter	Value
Design Wind Speed (Top)	39.0 m/s
Wind Base Shear (X)	76.7 kN
Wind Base Shear (Y)	92.0 kN

Concrete Material Breakdown (IS 10262:2019)

Material	Per m³	Total
Wet Concrete Volume		72.34 m³
Cement (50kg bags)	7.6 bags	549.8 bags
Fine Aggregate (Sand)	0.419 m³	30.29 m³
Coarse Aggregate (20mm)	0.821 m³	59.37 m³
Water	171 kg	12371 kg
Water:Cement Ratio	0.45	—
Nominal Mix Ratio	1:1.76:3.13	—

Note: Indicative quantities per IS 10262:2019. Add 3-5% wastage for actual procurement.

Beam L/d Deflection Check (IS 456 Cl 23.2)

Beam ID	Span (mm)	Depth (mm)	Actual L/d	Allowable	Status
AB_V_1_L0	4000	350	11.4	20	PASS
AB_V_2_L0	2000	300	6.7	20	PASS
AB_V_3_L0	2000	300	6.7	20	PASS
AB_V_4_L0	4000	350	11.4	20	PASS
AB_V_5_L0	2000	300	6.7	20	PASS
AB_V_6_L0	3000	300	10.0	20	PASS
AB_V_7_L0	4000	350	11.4	20	PASS
AB_V_8_L0	4000	350	11.4	20	PASS
AB_V_9_L0	1000	300	3.3	20	PASS
AB_V_10_L0	3000	300	10.0	20	PASS
AB_H_11_L0	5000	450	11.1	20	PASS

Beam ID	Span (mm)	Depth (mm)	Actual L/d	Allowable	Status
AB_H_12_L0	5000	450	11.1	20	PASS
AB_H_13_L0	2500	300	8.3	20	PASS
AB_H_14_L0	3750	350	10.7	20	PASS
AB_H_15_L0	3750	350	10.7	20	PASS
AB_H_16_L0	2500	300	8.3	20	PASS
AB_H_17_L0	5000	450	11.1	20	PASS
AB_H_18_L0	5000	450	11.1	20	PASS
AB_V_1_L1	4000	350	11.4	20	PASS
AB_V_2_L1	2000	300	6.7	20	PASS
AB_V_3_L1	2000	300	6.7	20	PASS
AB_V_4_L1	4000	350	11.4	20	PASS
AB_V_5_L1	2000	300	6.7	20	PASS
AB_V_6_L1	3000	300	10.0	20	PASS
AB_V_7_L1	4000	350	11.4	20	PASS
AB_V_8_L1	4000	350	11.4	20	PASS
AB_V_9_L1	1000	300	3.3	20	PASS
AB_V_10_L1	3000	300	10.0	20	PASS
AB_H_11_L1	5000	450	11.1	20	PASS
AB_H_12_L1	5000	450	11.1	20	PASS
AB_H_13_L1	2500	300	8.3	20	PASS
AB_H_14_L1	3750	350	10.7	20	PASS
AB_H_15_L1	3750	350	10.7	20	PASS
AB_H_16_L1	2500	300	8.3	20	PASS
AB_H_17_L1	5000	450	11.1	20	PASS
AB_H_18_L1	5000	450	11.1	20	PASS

Stability & Fire Resistance (IS 456 Table 16A)

Members Checked: 72 | Passed Stability: 72/72 | Passed Fire: 72/72

Code Compliance Audit

Check	Status	Notes
Column Axial Capacity	PASS	Load: 208.7 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 193.5 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 188.4 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 178.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 107.2 vs Cap: 1781.2 kN

Check	Status	Notes
Column Axial Capacity	PASS	Load: 136.9 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 107.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 117.3 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 213.1 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 187.7 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 213.1 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 91.9 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 91.7 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 66.5 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 91.7 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 137.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 168.1 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 107.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 71.1 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 63.5 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 60.9 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 55.9 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 53.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 57.1 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 53.6 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 58.7 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 95.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 82.5 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 95.2 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 46.0 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 42.1 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 33.3 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 42.1 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 68.8 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 84.0 vs Cap: 1781.2 kN
Column Axial Capacity	PASS	Load: 53.6 vs Cap: 1781.2 kN
Bearing Pressure	PASS	Press: 68.0 <= 200.0
Punching Shear	PASS	Shear Stress: 0.39 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 63.1 <= 200.0
Punching Shear	PASS	Shear Stress: 0.36 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 61.4 <= 200.0
Punching Shear	PASS	Shear Stress: 0.35 <= 1.25 N/mm2

Check	Status	Notes
Bearing Pressure	PASS	Press: 58.1 <= 200.0
Punching Shear	PASS	Shear Stress: 0.33 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 34.9 <= 200.0
Punching Shear	PASS	Shear Stress: 0.20 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 44.6 <= 200.0
Punching Shear	PASS	Shear Stress: 0.25 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 34.9 <= 200.0
Punching Shear	PASS	Shear Stress: 0.20 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 38.2 <= 200.0
Punching Shear	PASS	Shear Stress: 0.22 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 69.5 <= 200.0
Punching Shear	PASS	Shear Stress: 0.39 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 61.2 <= 200.0
Punching Shear	PASS	Shear Stress: 0.35 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 69.5 <= 200.0
Punching Shear	PASS	Shear Stress: 0.39 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 30.0 <= 200.0
Punching Shear	PASS	Shear Stress: 0.17 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 29.9 <= 200.0
Punching Shear	PASS	Shear Stress: 0.17 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 21.7 <= 200.0
Punching Shear	PASS	Shear Stress: 0.12 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 29.9 <= 200.0
Punching Shear	PASS	Shear Stress: 0.17 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 44.9 <= 200.0
Punching Shear	PASS	Shear Stress: 0.25 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 54.8 <= 200.0
Punching Shear	PASS	Shear Stress: 0.31 <= 1.25 N/mm2
Bearing Pressure	PASS	Press: 34.9 <= 200.0
Punching Shear	PASS	Shear Stress: 0.20 <= 1.25 N/mm2
Beam Deflection (Serviceability)	PASS	Defl: 6.624mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 0.648mm vs Limit 8.00mm
Beam Deflection (Serviceability)	PASS	Defl: 0.648mm vs Limit 8.00mm
Beam Deflection (Serviceability)	PASS	Defl: 6.624mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 0.648mm vs Limit 8.00mm
Beam Deflection (Serviceability)	PASS	Defl: 3.281mm vs Limit 12.00mm
Beam Deflection (Serviceability)	PASS	Defl: 6.624mm vs Limit 16.00mm

Check	Status	Notes
Beam Deflection (Serviceability)	PASS	Defl: 6.624mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 0.041mm vs Limit 4.00mm
Beam Deflection (Serviceability)	PASS	Defl: 3.281mm vs Limit 12.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.823mm vs Limit 20.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.823mm vs Limit 20.00mm
Beam Deflection (Serviceability)	PASS	Defl: 1.582mm vs Limit 10.00mm
Beam Deflection (Serviceability)	PASS	Defl: 5.117mm vs Limit 15.00mm
Beam Deflection (Serviceability)	PASS	Defl: 5.117mm vs Limit 15.00mm
Beam Deflection (Serviceability)	PASS	Defl: 1.582mm vs Limit 10.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.823mm vs Limit 20.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.823mm vs Limit 20.00mm
Beam Deflection (Serviceability)	PASS	Defl: 6.624mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 0.648mm vs Limit 8.00mm
Beam Deflection (Serviceability)	PASS	Defl: 0.648mm vs Limit 8.00mm
Beam Deflection (Serviceability)	PASS	Defl: 6.624mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 0.648mm vs Limit 8.00mm
Beam Deflection (Serviceability)	PASS	Defl: 3.281mm vs Limit 12.00mm
Beam Deflection (Serviceability)	PASS	Defl: 6.624mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 6.624mm vs Limit 16.00mm
Beam Deflection (Serviceability)	PASS	Defl: 0.041mm vs Limit 4.00mm
Beam Deflection (Serviceability)	PASS	Defl: 3.281mm vs Limit 12.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.823mm vs Limit 20.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.823mm vs Limit 20.00mm
Beam Deflection (Serviceability)	PASS	Defl: 1.582mm vs Limit 10.00mm
Beam Deflection (Serviceability)	PASS	Defl: 5.117mm vs Limit 15.00mm
Beam Deflection (Serviceability)	PASS	Defl: 5.117mm vs Limit 15.00mm
Beam Deflection (Serviceability)	PASS	Defl: 1.582mm vs Limit 10.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.823mm vs Limit 20.00mm
Beam Deflection (Serviceability)	PASS	Defl: 7.823mm vs Limit 20.00mm

14. Professional Disclaimer

IMPORTANT NOTICE - READ CAREFULLY

This structural design report is generated by automated software for **PRELIMINARY DESIGN AND EDUCATIONAL PURPOSES ONLY**.

Key Limitations:

• The software uses simplified analysis methods (tributary area, short column assumptions).
• Seismic forces (IS 1893) and Wind loads (IS 875 Pt 3) have been incorporated for basic checks, but require full dynamic analysis for high-rise buildings.
• Site-specific soil conditions, groundwater, and other geotechnical factors are not evaluated.
• Local building codes may have additional requirements beyond IS 456:2000.

Required Actions Before Construction:
• All designs MUST be reviewed and sealed by a licensed Professional Engineer (PE).
• Independent structural analysis should be performed using appropriate software.
• Geotechnical investigation and foundation recommendations are required.
• All applicable permits and approvals must be obtained.

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All designs must be verified by a licensed Structural Engineer prior to construction.
Safety Factors: γc = 1.5 (concrete), γs = 1.15 (steel) | Load Factor: 1.5 (DL + LL)
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